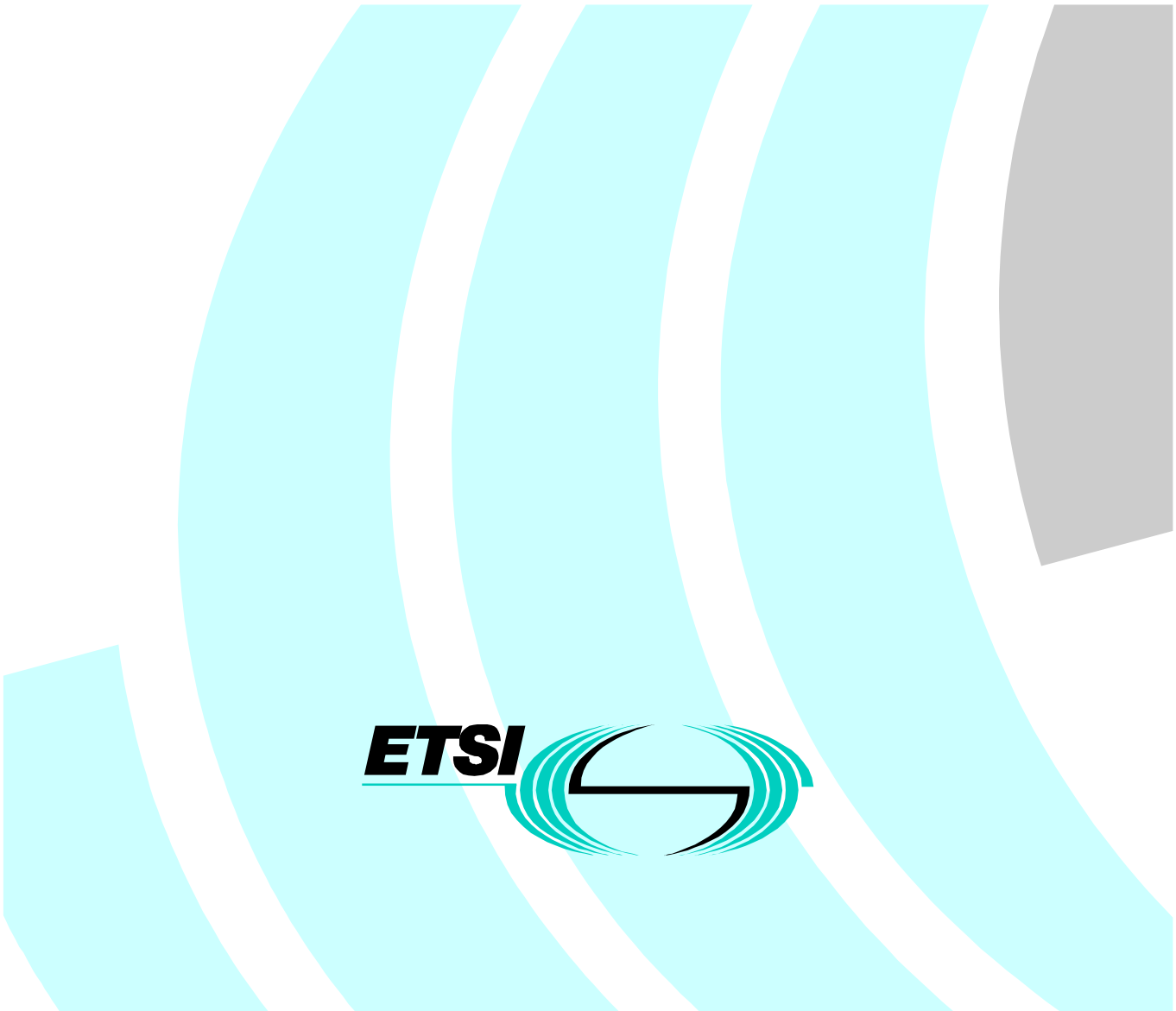
EN 300 058-5 V1.2.4 (1998-06)

European Standard (Telecommunications series)

Integrated Services Digital Network (ISDN); Call Waiting (CW) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 5: Test Suite Structure and Test Purposes (TSS&TP) specification for the network



Reference
REN/SPS-05145-F-5 (0jd90iqo.PDF)

Keywords
ISDN, DSS1, supplementary service, CW,
testing, TSS&TP, network

ETSI

Postal address
F-06921 Sophia Antipolis Cedex - FRANCE

Office address
650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16
Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Internet
secretariat@etsi.fr
http://www.etsi.fr
http://www.etsi.org

Copyright Notification

No part may be reproduced except as authorized by written permission.
The copyright and the foregoing restriction extend to reproduction in all media.

© European Telecommunications Standards Institute 1998.
All rights reserved.

Contents

Intellectual Property Rights 4
Foreword 4
1 Scope 5
2 Normative references 5
3 Definitions 6
3.1 Definitions related to conformance testing 6
3.2 Definitions related to EN 300 058-1 6
4 Abbreviations 7
5 Test Suite Structure (TSS) 7
6 Test Purposes (TP) 7
6.1 Introduction 7
6.1.1 TP naming convention 7
6.1.2 Source of TP definition 8
6.1.3 TP structure 8
6.1.4 Test strategy 8
6.1.5 Test of call states 9
6.2 Network TPs for CW 9
6.2.1 Network (S/T) 9
6.2.1.1 Called user interface 9
6.2.1.2 Calling user interface 11
6.2.2 Network (T) 12
6.2.2.1 Called user on a private ISDN 12
6.2.2.2 Calling user on a private ISDN 12
7 Compliance 12
8 Requirements for a comprehensive testing service 12
Annex A (informative): Changes with respect to the previous ETS 300 058-5 13
History 14

Intellectual Property Rights

IPRs essential or potentially essential to the present document may have been declared to ETSI. The information pertaining to these essential IPRs, if any, is publicly available for **ETSI members and non-members**, and can be found in SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available **free of charge** from the ETSI Secretariat. Latest updates are available on the ETSI Web server (http://www.etsi.fr/ipr or http://www.etsi.org/ipr).

Pursuant to the ETSI IPR Policy, no investigation, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Foreword

This European Standard (Telecommunications series) has been produced by ETSI Technical Committee Signalling Protocols and Switching (SPS).

The present document is part 5 of a multi-part standard covering the Digital Subscriber Signalling System No. one (DSS1) protocol specification for the Integrated Services Digital Network (ISDN) Call Waiting (CW) supplementary service, as described below:

Part 1: "Protocol specification";

Part 2: "Protocol Implementation Conformance Statement (PICS) proforma specification";

Part 3: "Test Suite Structure and Test Purposes (TSS&TP) specification for the user";

Part 4: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the user";

Part 5: "Test Suite Structure and Test Purposes (TSS&TP) specification for the network";

Part 6: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the network".

The present version updates the references to the basic call specifications.

National transposition dates	
Date of adoption of this EN:	19 June 1998
Date of latest announcement of this EN (doa):	30 September 1998
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	31 March 1999
Date of withdrawal of any conflicting National Standard (dow):	31 March 1999

1 Scope

This fifth part of EN 300 058 specifies the Test Suite Structure and Test Purposes (TSS&TP) for the Network side of the T reference point or coincident S and T reference point (as defined in ITU-T Recommendation I.411 [6]) of implementations conforming to the stage three standard for the Call Waiting (CW) supplementary service for the pan-European Integrated Services Digital Network (ISDN) by means of the Digital Subscriber Signalling System No. one (DSS1) protocol, EN 300 058-1 [1].

A further part of the present document specifies the Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma based on the present document. Other parts specify the TSS&TP and the ATS and partial PIXIT proforma for the User side of the T reference point or coincident S and T reference point of implementations conforming to EN 300 058-1 [1].

2 Normative references

References may be made to:

a) specific versions of publications (identified by date of publication, edition number, version number, etc.), in which case, subsequent revisions to the referenced document do not apply; or

b) all versions up to and including the identified version (identified by "up to and including" before the version identity); or

c) all versions subsequent to and including the identified version (identified by "onwards" following the version identity); or

d) publications without mention of a specific version, in which case the latest version applies.

A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] EN 300 058-1 (V1.2): "Integrated Services Digital Network (ISDN); Call Waiting (CW) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[2] EN 300 058-2 (V1.2): "Integrated Services Digital Network (ISDN); Call Waiting (CW) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 2: Protocol Implementation Conformance Statement (PICS) proforma specification".

[3] ISO/IEC 9646-1: "Information technology - OSI Conformance Testing Methodology and Framework; Part 1: General Concepts".

[4] ISO/IEC 9646-2: "Information technology - OSI Conformance Testing Methodology and Framework; Part 2: Abstract Test Suite specification".

[5] ISO/IEC 9646-3: "Information technology - OSI Conformance Testing Methodology and Framework; Part 3: The Tree and Tabular Combined Notation".

[6] ITU-T Recommendation I.411 (1993): "ISDN user-network interfaces - Reference configurations".

[7] EN 300 403-1: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Signalling network layer for circuit-mode basic call control; Part 1: Protocol specification [ITU-T Recommendation Q.931 (1993), modified]".

[8] ITU-T Recommendation I.112: "Vocabulary and terms for ISDNs".

[9] CCITT Recommendation E.164: "Numbering plan for the ISDN era".

[10] ITU-T Recommendation I.210: "Principles of the telecommunication services supported by an ISDN and the means to describe them".

[11] EN 300 403-3: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Signalling network layer for circuit-mode basic call control; Part 3: Protocol Implementation Conformance Statement (PICS) proforma specification".

3 Definitions

For the purposes of the present document, the following definitions apply:

3.1 Definitions related to conformance testing

abstract test case: Refer to ISO/IEC 9646-1 [3].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [3].

active test: A test case where the IUT is required to send a particular message, but not in reaction to a received message. This would usually involve the use of PIXIT information to see how this message can be generated and quite often is specified in an ATS using an implicit send event.

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [3].

implicit send event: Refer to ISO/IEC 9646-3 [5].

lower tester: Refer to ISO/IEC 9646-1 [3].

passive test: A test case where the IUT is required to respond to a protocol event (e.g. received message) with another protocol event (e.g. send message) which normally does not require any special operator intervention as associated with the implicit send event.

point of control and observation: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation Conformance Statement (PICS): Refer to ISO/IEC 9646-1 [3].

PICS proforma: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation Extra Information For Testing (PIXIT): Refer to ISO/IEC 9646-1 [3].

PIXIT proforma: Refer to ISO/IEC 9646-1 [3].

system under test: Refer to ISO/IEC 9646-1 [3].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [3].

3.2 Definitions related to EN 300 058-1

Integrated Services Digital Network (ISDN): See ITU-T Recommendation I.112 [8], definition 308.

ISDN number: A number conforming to the numbering and structure specified in CCITT Recommendation E.164 [9].

network: The DSS1 protocol entity at the Network side of the user-network interface where a T reference point or coincident S and T reference point applies.

network (S/T): The DSS1 protocol entity at the Network side of the user-network interface where a coincident S and T reference point applies.

network (T): The DSS1 protocol entity at the Network side of the user-network interface where a T reference point applies (Network connected to Private ISDN).

service; telecommunication service: See ITU-T Recommendation I.112 [8], definition 201.

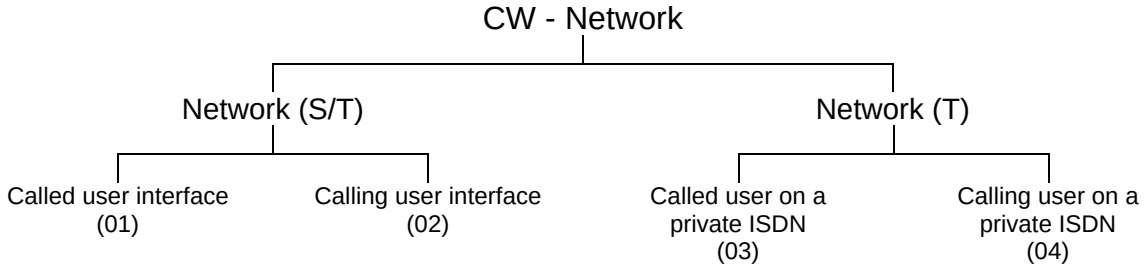
supplementary service: See ITU-T Recommendation I.210 [10], subclause 2.4.

4 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ATM	Abstract Test Method
ATS	Abstract Test Suite
CR1	Call Reference 1
CR2	Call Reference 2
CW	Call Waiting
ISDN	Integrated Services Digital Network
IUT	Implementation Under Test
N00	Null call state
N03	Outgoing Call Proceeding call state
N04	Call Delivered call state
N06	Call Present call state
N07	Call Received call state
N09	Incoming Call Proceeding call state
N10	Active call state
N12	Disconnect Indication call state
N19	Release Request call state
N25	Overlap Receiving call state
PICS	Protocol Implementation Conformance Statement
PIXIT	Protocol Implementation eXtra Information for Testing
TP	Test Purpose
TSS	Test Suite Structure

5 Test Suite Structure (TSS)



NOTE: Numbers in brackets represent group numbers and are used in TP identifiers.

Figure 1: Test suite structure

6 Test Purposes (TP)

6.1 Introduction

For each test requirement a TP is defined.

6.1.1 TP naming convention

The TPs are numbered, starting at 001, within each group. Groups are organized according to the TSS. Additional references are added to identify the actual test suite and whether it applies to the network or the user (see table 1).

Table 1: TP identifier naming convention scheme

Identifier: **<ss>_<iut><group>_<nnn>**				
<ss>	=	supplementary service:	e.g. "CW"	
<iut>	=	type of IUT:	U	User
			N	Network
<group>	=	group	2 digit field representing group reference according to TSS	
<nnn>	=	sequential number	(001-999)	

6.1.2 Source of TP definition

The TPs are based on EN 300 058-1 [1].

6.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used and this is illustrated in table 2. This table should be read in conjunction with any TP i.e. use a TP as an example to fully understand the table.

Table 2: Structure of a single TP

TP part	Text	Example
Header	<Identifier> *tab* <paragraph number in base ETS> *tab* <condition> *CR*.	see table 1 subclause 0.0.0 mandatory, optional
Stimulus	Ensure that the IUT in the <basic call state> <trigger> *see below for message structure* *or* <goal>	 N00, N10, etc. receiving a XXXX message to request a ...
Reaction	<action> <conditions> *if the action is sending* *see below for message structure* <next action>, *etc.* and enters <supplementary service state> *and/or* and remains in the same state(s) *or* and enters state <state> with CR<number(s)>	sends, saves, does, etc. using en bloc sending, ...
Message structure	<message type> message containing a a) <info element> information element with b) a <field name> encoded as *or* including <coding of the field> and *back to a or b,*	SETUP, FACILITY, CONNECT, ... Bearer capability, Facility, ...
NOTE: Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.		

6.1.4 Test strategy

As the base standard EN 300 058-1 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification EN 300 058-2 [2]. The criteria applied include the following:

- only the requirements from the point of view of the T or coincident S and T reference point are considered;
- whether or not a test case can be built from the TP is not considered.

6.1.5 Test of call states

Many TPs include a reference to the IUT's final call state after the realization of the TP. In these cases the TP includes the requirement to ensure that the IUT has entered this particular final call state. Ensuring that the IUT is in a particular call state shall be realized by following the procedures described in subclause 5.8.10 of EN 300 403-1 [7]. According to these procedures, the IUT on receipt of a STATUS ENQUIRY message, shall respond with a STATUS message indicating, in the third octet of the Call state information element, the current call state of the IUT. This exchange of messages is not mentioned explicitly in each TP but is considered to be implicit in the reference to the final call state. This way of phrasing the TPs has been used to avoid over-complicating the text and structure of the TPs and to improve the readability.

6.2 Network TPs for CW

All PICS items referred to in this subclause are as specified in EN 300 058-2 [2] unless indicated otherwise by another numbered reference.

The following convention is used:

CR1: The call which is in the Active call state when the second call arrives at the access;
CR2: The second call which arrives at the access encountering a channels busy condition.

6.2.1 Network (S/T)

Selection: IUT supports coincident S and T reference point procedures. PICS: R 3.1.

6.2.1.1 Called user interface

CW_N01_001 subclause 9.4.1 mandatory
Ensure that the IUT in the Active call state N10 for the call CR1 and in the Null call state N00 for the call CR2, if a B-channel is not available, to indicate an incoming call,
sends a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel" and the preferred/exclusive bit set to "indicated channel is preferred", and enters state N06 for the call CR2.

CW_N01_002 subclause 9.5.1 mandatory
Ensure that the IUT in the Active call state N10 for the call CR1 and in the Overlap Receiving call state N25 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of an ALERTING message for CR2,
sends no message and enters state N07 for the call CR2.

CW_N01_003 subclause 9.5.1 mandatory
Ensure that the IUT in the Active call state N10 for the call CR1 and in the Call Present call state N06 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of an ALERTING message for CR2,
sends no message and enters state N07 for the call CR2.

CW_N01_004 subclause 9.5.1 mandatory
Ensure that the IUT in the Active call state N10 for the call CR1 and in the Incoming Call Proceeding call state N09 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of an ALERTING message for CR2,
sends no message and enters state N07 for the call CR2.

CW_N01_005 subclause 9.6.1 mandatory
Ensure that the IUT in the Null call state N00 for the call CR1 which has just been released and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of a CONNECT message for CR2,
sends a CONNECT ACKNOWLEDGE message for CR2 with a Channel identification information element coded as "channel is indicated, no alternative acceptable" in the information channel selection field and enters state N10 for the call CR2.

CW_N01_006 subclause 9.6.1 mandatory

Ensure that the IUT in the Active call state N10 and the Call Held auxiliary state for CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of a CONNECT message for CR2,

sends a CONNECT ACKNOWLEDGE message for CR2 with a Channel identification information element coded as "channel is indicated, no alternative acceptable" in the information channel selection field and enters state N10 for the call CR2.

CW_N01_007 subclause 9.6.2 mandatory

Ensure that the IUT in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of a DISCONNECT message for CR2,

sends a RELEASE message for CR2 and enters state N19 for the call CR2.

CW_N01_008 subclause 9.5.2, 9.6.2 optional

Ensure that the IUT (with a point to multipoint configuration) in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on expiry of timer T-CW,

sends a RELEASE message for CR2 with cause #102 "recovery on timer expiry" and enters state N19 for the call CR2.

Selection: IUT supports timer T-CW. PICS: TM2.
Selection: IUT supports basic access, point-to-multipoint configuration? PICS: EN 300 403-3 [11] MC 2.5.

CW_N01_009 subclause 9.5.2, 9.6.2 optional

Ensure that the IUT (with a point to point configuration) in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on expiry of timer T-CW,

sends a DISCONNECT message for CR2 with cause #102 "recovery on timer expiry" and enters state N12 for the call CR2.

Selection: IUT supports timer T-CW. PICS: TM2.
Selection: IUT supports primary rate access OR basic access, point-to-point configuration? PICS: EN 300 403-3 [11] MC 2.4.

CW_N01_010 subclause 9.5.2, 9.6.2 optional

Ensure that the IUT (with a point to multipoint configuration) in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on expiry of timer T301,

sends a DISCONNECT for CR2 message and enters state N12 for the call CR2.

Selection: IUT does not support timer T-CW. PICS: NOT TM 2 AND IUT supports timer T301. PICS: EN 300 403-3 [11] TM 1.
Selection: IUT supports basic access, point-to-multipoint configuration? PICS: EN 300 403-3 [11] MC 2.5.

CW_N01_011 subclause 9.5.2, 9.6.2 optional

Ensure that the IUT (with a point to point configuration) in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on expiry of timer T301,

sends a DISCONNECT for CR2 message and enters state N12 for the call CR2.

Selection: IUT does not support timer T-CW. PICS: NOT TM 2 AND IUT supports timer T301. PICS: EN 300 403-3 [11] TM 1.
Selection: IUT supports primary rate access OR basic access, point-to-point configuration? PICS: EN 300 403-3 [11] MC 2.4.

CW_N01_012 subclause 9.6.2 mandatory
Ensure that the IUT in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message containing a Channel identification information element with the information channel selection field coded as "no channel", to indicate that the remote user has cleared the call,
sends a DISCONNECT for CR2 message and enters state N12 for the call CR2.

CW_N01_013 subclause 9.6.2 mandatory
Ensure that the IUT in the Active call state N10 for the call CR1 and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of a CONNECT message for CR2 and if no B-channel is available to complete the call,
sends a RELEASE message for CR2 with cause #44 "requested circuit/channel not available" and enters state N19 for the call CR2.

NOTE: The call CR1 is still in the active call state so that no B-channel is available for the call CR2.

CW_N01_014 subclause 9.6.2 mandatory
Ensure that the IUT in the Null call state N00 for the call CR1 which has just been released and in the Call Received call state N07 for the call CR2, after initially sending a SETUP message (CR2) containing a Channel identification information element with the information channel selection field coded as "no channel", on receipt of a CONNECT message for CR2 with a Channel identification information element coded as "channel is indicated, no alternative acceptable" and the IUT cannot accept the indicated B-channel,
sends a RELEASE message with cause #44 "requested circuit/channel not available" and enters state N19 for the call CR2.

6.2.1.2 Calling user interface

CW_N02_001 subclause 9.5.2 optional
Ensure that the IUT in the Call Delivered call state N04, to indicate the expiry of timer T-CW at the called user side,
sends a DISCONNECT message with cause #19 "no answer from user (user alerted)" and enters state N12.

CW_N02_002 subclause 9.5.1.1, 10.1 optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03, to indicate that the called user has sent an ALERTING message (where the call is a waiting call),
sends an ALERTING message containing a Notification indicator information element coded as "call is a waiting call" and enters state N04.

Selection: IUT supports inclusion of Notification indicator information element in ALERTING message. PICS: MC 10.

6.2.2 Network (T)

Selection: IUT supports T reference point procedures. PICS: R 3.2.

6.2.2.1 Called user on a private ISDN

CW_N03_001 subclause 10.1 mandatory
Ensure that the IUT in the Incoming Call Proceeding call state N09, on receipt of an ALERTING message, containing a Notification indicator information element coded as "call is a waiting call",
sends no message and enters state N07.

6.2.2.2 Calling user on a private ISDN

CW_N04_001 subclause 10.2 optional
Ensure that the IUT in the Outgoing Call Proceeding call state N03, to indicate that the called user has sent an ALERTING message (where the call is a waiting call),
sends an ALERTING message containing a Notification indicator information element coded as "call is a waiting call" and enters state N04.

Selection: IUT supports inclusion of Notification indicator information element in ALERTING message sent to private ISDN. PICS: MC 10.

7 Compliance

An ATS which complies with this TSS&TP specification shall:

a) consist of a set of test cases corresponding to the set or to a subset of the TPs specified in clause 6;

b) use a TSS which is an appropriate subset of the whole of the TSS specified in clause 5;

c) use the same naming conventions for the test groups and test cases;

d) maintain the relationship specified in clause 6 between the test groups and TPs and the entries in the PICS proforma to be used for test case deselection;

e) comply with ISO/IEC 9646-2 [4].

In the case of a) or b) above, a subset shall be used only where a particular Abstract Test Method (ATM) makes some TPs untestable. All testable TPs from clause 6 shall be included in a compliant ATS.

8 Requirements for a comprehensive testing service

As a minimum the Remote test method, as specified in ISO/IEC 9646-2 [4], shall be used by any organization claiming to provide a comprehensive testing service for network equipment claiming conformance to EN 300 058-1 [1].

Annex A (informative): Changes with respect to the previous ETS 300 058-5

The following changes have been done:

- conversion to EN layout;
- replacement of references to ETS 300 102 with EN 300 403;
- replacement of references to I-ETSs with EN 300 403;
- substitution of non-specific references to basic standards where the intention is to refer to the latest version.

History

Document history		
Edition 1	May 1997	Publication as ETS 300 058-5
V1.2.3	February 1998	One-step Approval Procedure OAP 9824: 1998-02-13 to 1998-06-12
V1.2.4	June 1998	Publication

ISBN 2-7437-2241-X
Dépôt légal : Juin 1998